



Safety Requirements Specification (SRS)

Hybrit Pilot Plant

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Table of Contents

1	Purpose and Scope	4
1.1	General Project Information	4
1.2	Standards Used	4
2	General SIS Requirements	5
2.1	SIS Logic Solver Hardware Requirements	5
2.2	SIS Application Software Requirements.....	6
3	General SIF Requirements	8
4	Safety Instrumented Functions	9
4.1	SIF List	9
4.1.1	Low Demand Service SIFs	9
5	SIF-058 SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningsenhet (345EC015).....	12
6	SIF-059 SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566)	15
7	SIF-060 SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tillopp spolvatten (345SV634)	18
8	SIF-061 SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566)	21
9	SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV634) ...	24
10	SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)	27
11	SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)	30
12	SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)	33
13	SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635).....	36
14	SIF-067 SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högspänningsenhet (353EC015)	39
15	SIF-068 SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566).....	42
16	SIF-069 SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635) ..	45
17	SIF-070 SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566)	48
18	SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635) 51	
19	SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)	54
20	SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)	57
21	SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)	60
22	Abbreviations and Definitions	63
22.1	Abbreviations	63
22.2	Definitions	63
23	Disclaimer, Assumptions, Equipment Data	64
23.1	Disclaimer.....	64
23.2	Assumptions Safety Requirements Specification.....	64

1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the general and specific safety requirements specification (SRS) for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project in accordance with clause 10 of IEC 61511-1 / ISA 61511-1. This document provides functional requirements that are common to the SIS and all of its associated SIFs, as well as the functional and integrity requirements that are unique to each SIF. In this way repetition, conflicting, or missing requirements are reduced.

The Safety Instrumented System needs to achieve the following key functional objectives:

- Provide reliable protection for the plant personnel, the environment, and process equipment from adverse effects due to process upsets
- Provide protection against spurious / false trips to minimize lost production, equipment damage, and exposure to process hazards
- Provide effective and reliable means of routine inspection and testing of the SIS

The Safety Instrumented System includes all elements from the sensor to the final element including logic solver inputs, outputs, communications, and power supplies.

This document does not cover requirements for the following:

- Basic Process Control System (BPCS)
- Non-SIS Independent Layers of Protection

1.1 General Project Information

Project Identification: 001
Project Name: Hybrit Pilot Plant
Project Description:

1.2 Standards Used

This SRS is based on the following standards.

DOCUMENT NUMBER - TITLE	REVISION	REVISION DATE	TYPE
IEC 61508 - Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems	2010	4/30/2010	
IEC 61511 - Functional safety: Safety Instrumented Systems for the Process Industry Sector	2017	8/31/2017	
ANSI/ISA 84.00.01 - Functional safety: Safety Instrumented Systems for the Process Industry Sector	2004	9/30/2004	

2 General SIS Requirements

This chapter describes the requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project. Additional or different requirements for individual SIFs will be documented in the appropriate SIF subsection of this document.

2.1 SIS Logic Solver Hardware Requirements

This section describes the Logic Solver hardware requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project.

SYSTEMATIC FAILURE PROTECTION	
Logic Solver SIL Capability (Systematic)	3
SIS LOGIC SOLVER FAILURE REQUIREMENTS	
Action on Logic Solver Fault Detection	Vote channel to trip
Actions to Achieve/Maintain Safe State when SIS is Degraded or Inoperative	
Logic Solver Mean Time to Fail Spurious	20
SIS Output State Combinations to Avoid	None
SIS INTERFACES TO ANY OTHER SYSTEMS	
Engineering Station	Engineering Station must be implemented in accordance with the project guidelines
BPCS	Control System must be implemented in accordance with the project guidelines
Operator Station (HMI)	Operator Stations must be implemented in accordance with the project guidelines
Historian/Sequence of Events	
Other	
SIS FUNCTIONAL REQUIREMENTS	
Independent Manual Shutdown	A pushbutton will be provided in the control room to manually initiate shutdown
Major Accident Survivability	2 hours
OPERATIONAL REQUIREMENTS	
SIS Start-up / Restart Requirements	None
LOGIC SOLVER ENVIRONMENTAL EXTREMES	
Ambient Temperature	0-40 °C
Humidity	0-80%
Electromagnetic Interference	0-10 V/m
Radio Interference	3 V/m
Electrostatic Discharge	6 kV
Area Classification	Logic solver placed outside atex zone
Lightning Protection	
Other	
NOTES	

USER DEFINED REQUIREMENTS	

2.2 SIS Application Software Requirements

This section describes the application software requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project.

APPLICATION PROGRAM			
General	All configuration and programming requirements specified by the safety manual for the IEC 61508 SIL 3 capable PLC shall be followed.		
Program Sequencing	Inputs blocks should precede logic functions, output blocks should be last		
Trip/Time Delays	None		
Process Modes of Operation	The following operating modes are identified for each Safety Instrumented Functions: startup, normal operation, shutdown, failed and degraded		
REAL TIME PERFORMANCE REQUIREMENTS			
CPU Capacity	x.x GHz		
Network Bandwidth	x Mbs		
Acceptable Performance with Faults			
Specified Time all Trip Signals Received	0m 0s 050ms		
PROCESS VARIABLE VALIDATION			
Process Variable (PV) Validation Criteria	Range	Low	High
	Analog Signals	Under-range	Over-range
		< 3.6 mA	> 21.5 mA
Action on PV Out of Range	Vote for trip		
Action on PV Excessive Rate of Change			
Action on PV Frozen Value	Vote for trip		
Action on PV Open Circuit Detection	Vote for trip		
Action on PV Short Circuit Detection	Vote for trip		
MONITORING FUNCTIONALITY			
Application Program Self-Monitoring	Included in SIL 3 capable PLC programming environment		
Device Monitoring	Implement line monitoring on all Energize to Trip functions		
PROOF TESTING REQUIREMENTS			
Proof Testing while Process is Operational	No		
Additional Field Device Test Requirements	None expected, verify with device specific safety manual once devices are selected		
APPLICATION PROGRAM FAILURES			
Dangerous Process States Resulting from Application Program Failure	Instability to perform safety function task, addressed by using SIL capable PLC		
COMMUNICATION INTERFACES			
Validation of Received Data	Bounds checking/range checking		

Validation of Transmitter Data	N/A, data send from SIL3 capable PLC
Measures to Limit Interface Use	N/A
REFERENCE DOCUMENTS	
Reference Type	Document Number - Title
	-
NOTES	
USER DEFINED REQUIREMENTS	

3 General SIF Requirements

This chapter describes the common functional requirements that generally apply to all SIFs in the Hybrit Pilot Plant project and that are not repeated for each specific SIF. Additional requirements for an individual SIF will be documented in the appropriate SIF subsection of this document.

Maintenance Override Degradation Requirements							
Voting Degradation	Original	1oo1	1oo2	2oo2	1oo3	2oo3	3oo3
	Degraded	Trip	1oo1	1oo1	1oo2	1oo2	2oo2
Field Equipment Fail Safe States							
Transmitter	Low output, e.g. 3.6 mA						
Switch	Open contacts, e.g. 0 mA						
Isolation Valve	Closed						
Vent Valve	Open						
Motor	Stop						
Relay	Open						
Reference Documents							
Reference Type		Document Number - Title					
		-					
Notes							

4 Safety Instrumented Functions

The functional and integrity requirements unique to each SIF are detailed in separate subsequent sections. Each requirement is uniquely numbered so it can be traced through project development from design, through implementation to operation.

4.1 SIF List

The following Safety Instrumented Functions are documented in this Safety Requirements Specification.

4.1.1 Low Demand Service SIFs

SIF TAG	REQUIRED		SIF NAME
	SIL	RRF	
SIF-058	1	10	SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningssenheter (345EC015) - SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningssenheter (345EC015)
SIF-059	2	100	SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566) - SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566)
SIF-060	2	100	SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tilllopp spolvatten (345SV634) - SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tilllopp spolvatten (345SV634)
SIF-061	2	100	SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566) - SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566)
SIF-062	2	100	SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tilllopp spolvatten (345SV634) - SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tilllopp spolvatten (345SV634)
SIF-063	2	100	SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522) - SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)
SIF-064	2	100	SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521) - SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)

SIF-065	2	100	SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520) - SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)
SIF-066	2	100	SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635) - SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)
SIF-067	1	10	SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högspänningsenhet (353EC015) - SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högspänningsenhet (353EC015)
SIF-068	2	0	SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566) - SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566)
SIF-069	2	0	SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635) - SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635)
SIF-070	2	0	SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566) - SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566)
SIF-071	2	0	SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635) - SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635)
SIF-072	2	0	SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522) - SIL2 Position avstängning venturihals (353SV520 &

			353SV521) => Open avstängning venturihals(353SV522)
SIF-073	2	0	SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521) - SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)
SIF-074	2	0	SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520) - SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)

5 SIF-058 SIL1 Syrehalt gasväg (345AZ362 | H2) => Trip Högsäpänningsenhet (345EC015)

SIF IDENTIFICATION								
SIF Tag			SIF-058		Revision		0	
SIF Name			SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningsenhet (345EC015)					
SIF Description / Service			SIL1 Syrehalt gasväg (345AZ362 H2) => Trip Högspänningsenhet (345EC015)					
FUNCTIONAL REQUIREMENTS								
Equipment			345AZ362 345EC015					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part			Sensor		Logic Solver		Final Element	
Response Time			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345AZ362	0	0	1.25	High	0	% O2	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345EC015	Open/Off		No	Open		No		345EC015_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

6 SIF-059 SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566)

SIF IDENTIFICATION								
SIF Tag		SIF-059		Revision		0		
SIF Name		SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566)						
SIF Description / Service		SIL2 Vätsketryck nedströms pump (345PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(345SV566)						
FUNCTIONAL REQUIREMENTS								
Equipment		345PI326 320PZ320/1/2 345SV566						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345PI326 < (320PZ320/1/2 + 0,5 bar)	0	0	0.5	Low	0	Bar	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV566	Open/Off		No	Closed		Yes		345SV566_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

7 SIF-060 SIL2 Vätskenivå sludge tank (345LS310|H2) => Close Avstängning tillopp spolvatten (345SV634)

SIF IDENTIFICATION								
SIF Tag		SIF-060		Revision		0		
SIF Name		SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tillopp spolvatten (345SV634)						
SIF Description / Service		SIL2 Vätskenivå sludge tank (345LS310 H2) => Close Avstängning tillopp spolvatten (345SV634)						
FUNCTIONAL REQUIREMENTS								
Equipment		345LS310 345SV634						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345LS310	0	0	0	Low	0	Bar(g)	Process	switch (nivå ej fastställd)
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV634	Close / On		No	Closed		Yes		345SV634_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

8 SIF-061 SIL2 Vätskenivå sludge tank (345LS312 | L2) => Close Avblödning slamvatten (345SV566)

SIF IDENTIFICATION								
SIF Tag		SIF-061			Revision		0	
SIF Name		SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566)						
SIF Description / Service		SIL2 Vätskenivå sludge tank (345LS312 L2) => Close Avblödning slamvatten (345SV566)						
FUNCTIONAL REQUIREMENTS								
Equipment		345LS312 345SV566						
Process Safe State					Process Safety Time		0h 0m 30s	
Demand Source								
Demand Rate		0.000E+000 per year			Demand Mode		Low	
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual			Local/Remote		Local	
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part		Sensor			Logic Solver		Final Element	
Response Time		msec			msec		msec	
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345LS312	0	0	0	Low	0	%	Process	switch (nivå ej fastställd)
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV566	Close / On		No	Closed		Yes		345SV566_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

9 SIF-062 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV634)

SIF IDENTIFICATION								
SIF Tag		SIF-062		Revision		0		
SIF Name		SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV634)						
SIF Description / Service		SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV634)						
FUNCTIONAL REQUIREMENTS								
Equipment		345LI354/5/6 345SV634						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345LI354/5/6	0	100	80	High	0	%	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV634	Close / On		No	Closed		No		345SV634_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

10 SIF-063 SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)

SIF IDENTIFICATION								
SIF Tag		SIF-063		Revision		0		
SIF Name		SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)						
SIF Description / Service		SIL2 Position avstängning venturihals (345SV520 & 345SV521) => Open avstängning venturihals(345SV522)						
FUNCTIONAL REQUIREMENTS								
Equipment		345SV520 & 345SV521 345SV522						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345SV520 & 345SV521	0	0	0	None	0	ej_SGO & ej_SGO	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV522	Open/Off		No	Open		Yes		345SV522_GSO

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

11 SIF-064 SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)

SIF IDENTIFICATION								
SIF Tag		SIF-064		Revision		0		
SIF Name		SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)						
SIF Description / Service		SIL2 Position avstängning venturihals (345SV520 & 345SV522) => Open avstängning venturihals(345SV521)						
FUNCTIONAL REQUIREMENTS								
Equipment		345SV520 & 345SV522 345SV521						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345SV520 & 345SV522	0	0	0	None	0	ej_SGO & ej_SGO	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV521	Open/Off		No	Open		Yes		345SV521_GSO

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

12 SIF-065 SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)

SIF IDENTIFICATION								
SIF Tag		SIF-065		Revision		0		
SIF Name		SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)						
SIF Description / Service		SIL2 Position avstängning venturihals (345SV521 & 345SV522) => Open avstängning venturihals(345SV520)						
FUNCTIONAL REQUIREMENTS								
Equipment		345SV521 & 345SV522 345SV520						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
345SV521 & 345SV522	0	0	0	None	0	ej_SGO & ej_SGO	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV520	Open/Off		No	Open		Yes		345SV520_GSO

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

13 SIF-066 SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)

SIF IDENTIFICATION								
SIF Tag		SIF-066		Revision		0		
SIF Name		SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)						
SIF Description / Service		SIL2 Tryckmätare hydrofor ((345PI420/1)< (320PZ320/1/2 + 0,5 bar)) => Close Avstängning tillopp spolvatten (345SV634) Close Avstängning tillopp spolvatten (345SV635)						
FUNCTIONAL REQUIREMENTS								
Equipment		345PI420/1 320PZ320/1/2 345SV634 345SV635						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
((345PI420/1)< (320PZ320/1/2 + 0,5 bar))	0	0	1	Low	0	Bar	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV634	Open/Off		No	Open		No		345SV634_GSS
345SV635	Open/Off		No	Open		No		345SV635_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

14 SIF-067 SIL1 Syrehalt gasväg (353AZ362 | H2) => Trip Högsäpänningsenhet (353EC015)

SIF IDENTIFICATION								
SIF Tag			SIF-067		Revision		0	
SIF Name			SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högspänningsenhet (353EC015)					
SIF Description / Service			SIL1 Syrehalt gasväg (353AZ362 H2) => Trip Högspänningsenhet (353EC015)					
FUNCTIONAL REQUIREMENTS								
Equipment			353AZ362 353EC015					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part			Sensor		Logic Solver		Final Element	
Response Time			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
353AZ362	0	0	1.25	High	0	% O2	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
353EC015	Open/Off		No	Open		No		353EC015_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

15 SIF-068 SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566)

SIF IDENTIFICATION								
SIF Tag		SIF-068			Revision		0	
SIF Name		SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566)						
SIF Description / Service		SIL2 Vätsketryck nedströms pump (353PI326 < (320PZ320/1/2 + 0,5 bar)) => Close Avblödning slamvatten(353SV566)						
FUNCTIONAL REQUIREMENTS								
Equipment		353PI326 320PZ320/1/2 353SV566						
Process Safe State					Process Safety Time		0h 0m 30s	
Demand Source								
Demand Rate		0.000E+000 per year			Demand Mode		Low	
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual			Local/Remote		Local	
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor			Logic Solver		Final Element	
		msec			msec		msec	
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
(353PI326 < (320PZ320/1/2 + 0,5 bar))	0	0	0.5	Low	0	Bar	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
353SV566	Open/Off		No	Open		No		353SV566_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

16 SIF-069 SIL2 Vätskenivå sludge tank (353LS310|H2) => Close Avstängning tillopp spolvatten (345SV635)

SIF IDENTIFICATION									
SIF Tag			SIF-069			Revision		0	
SIF Name			SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635)						
SIF Description / Service			SIL2 Vätskenivå sludge tank (353LS310 H2) => Close Avstängning tillopp spolvatten (345SV635)						
FUNCTIONAL REQUIREMENTS									
Equipment			353LS310 345SV635						
Process Safe State						Process Safety Time		0h 0m 30s	
Demand Source									
Demand Rate			0.000E+000 per year			Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown			Yes						
Protection Method			DeEnergizeToTrip						
Trip Reset			Manual			Local/Remote		Local	
Trip Reset Description									
Response Time SIF			10 seconds						
Maximum Part Response Time			Sensor			Logic Solver		Final Element	
			msec			msec		msec	
Mission Time			15 years						
INTENDED SENSOR LEGS									
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point	
353LS310	0	0	0	High	0	%	Process	switch (nivå ej fastställd)	
INTENDED FINAL ELEMENT LEGS									
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag	
345SV635	Close / On		No	Closed		Yes		345SV635_GSS	

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

17 SIF-070 SIL2 Vätskenivå sludge tank (353LS312 | L2) => Close Avblödning slamvatten (353SV566)

SIF IDENTIFICATION									
SIF Tag			SIF-070			Revision		0	
SIF Name			SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566)						
SIF Description / Service			SIL2 Vätskenivå sludge tank (353LS312 L2) => Close Avblödning slamvatten (353SV566)						
FUNCTIONAL REQUIREMENTS									
Equipment			345LS312 353SV566						
Process Safe State						Process Safety Time		0h 0m 30s	
Demand Source									
Demand Rate			0.000E+000 per year			Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown			Yes						
Protection Method			DeEnergizeToTrip						
Trip Reset			Manual			Local/Remote		Local	
Trip Reset Description									
Response Time SIF			10 seconds						
Maximum Part Response Time			Sensor			Logic Solver		Final Element	
			msec			msec		msec	
Mission Time			15 years						
INTENDED SENSOR LEGS									
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point	
353LS312	0	0	0	Low	0	%	Process	switch (nivå ej fastställd)	
INTENDED FINAL ELEMENT LEGS									
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag	
353SV566	Close / On		No	Closed		Yes		353SV566 _GSS	

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

18 SIF-071 SIL2 Hög vätskenivå WESP (353LI354/5/6|H2) => Close Avstängning tillopp spolvatten (345SV635)

SIF IDENTIFICATION								
SIF Tag		SIF-071		Revision		0		
SIF Name		SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635)						
SIF Description / Service		SIL2 Hög vätskenivå WESP (353LI354/5/6 H2) => Close Avstängning tillopp spolvatten (345SV635)						
FUNCTIONAL REQUIREMENTS								
Equipment		353LI354/5/6 345SV635						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
353LI354/5/6	0	0	80	High	0	%	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
345SV635	Close / On		No	Closed		Yes		345SV635_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

19 SIF-072 SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)

SIF IDENTIFICATION								
SIF Tag		SIF-072		Revision		0		
SIF Name		SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)						
SIF Description / Service		SIL2 Position avstängning venturihals (353SV520 & 353SV521) => Open avstängning venturihals(353SV522)						
FUNCTIONAL REQUIREMENTS								
Equipment		353SV520 & 353SV521 353SV522						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
353SV520 & 353SV521	0	0	0	None	0	ej_SGO & ej_SGO	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
353SV522	Open/Off		No	Open		Yes		353SV522_GSO

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

20 SIF-073 SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)

SIF IDENTIFICATION								
SIF Tag		SIF-073		Revision		0		
SIF Name		SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)						
SIF Description / Service		SIL2 Position avstängning venturihals (353SV520 & 353SV522) => Open avstängning venturihals(353SV521)						
FUNCTIONAL REQUIREMENTS								
Equipment		353SV520 & 353SV522 353SV521						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
353SV520 & 353SV522	0	0	0	None	0	ej_SGO & ej_SGO	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
353SV521	Open/Off		No	Open		Yes		353SV521_GSO

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

21 SIF-074 SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)

SIF IDENTIFICATION								
SIF Tag		SIF-074		Revision		0		
SIF Name		SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)						
SIF Description / Service		SIL2 Position avstängning venturihals (353SV521 & 353SV522) => Open avstängning venturihals(353SV520)						
FUNCTIONAL REQUIREMENTS								
Equipment		353SV521 & 353SV522 353SV520						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
353SV521 & 353SV522	0	0	0	None	0	ej_SGO & ej_SGO	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
353SV520	Open/Off		No	Open		Yes		353SV520_GSO

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	0
SIL Selection Method	Unknown		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

22 Abbreviations and Definitions

22.1 Abbreviations

BMS	Burner Management System
BPCS	Basic Process Control System
E/E/PE	Electrical/Electronic/Programmable Electronic
EMC	Electro-Magnetic Compatibility
FAT	Factory Acceptance Testing
FPL	Fixed Program Language
FVL	Full Variability Language
HFT	Hardware Fault Tolerance
HMI	Human Machine Interface
IEC	International Electrotechnical Commission
IPL	Independent Protection Layers
ISA	Instrumentation, Systems and Automation Society
LVL	Limited Variability Language
MoonN	"M" out of "N"
MTTFS	Mean Time To Fail Spurious
MRT	Mean Repair Time
PFD	Probability of Failure on Demand
PFDavg	Average Probability of Failure on Demand
PFH	Probability of a Dangerous Failure per Hour
PLC	Programmable Logic Controller
PTC	Proof Test Coverage
PTI	Proof Test Interval
RRF	Risk Reduction Factor
SAT	Site Acceptance Test
SERH	Safety Equipment Reliability Handbook
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SLC	Safety Life Cycle
SRS	Safety Requirements Specification

22.2 Definitions

Low Demand Mode	the demand is infrequent and the demand frequency is less than half the proof test frequency, e.g. proof tests are accounted for
High Demand Mode	the demand is somewhat frequent, the demand frequency is greater than half the proof test frequency, and the worst case diagnostic time interval is an order of magnitude smaller than the demand interval, e.g. proof tests are not accounted for, diagnostics are accounted for
Continuous Demand Mode	the demand is frequent, the demand frequency is greater than half the proof test frequency, and the worst case diagnostic time interval is not an order of magnitude smaller than the demand interval, e.g. proof tests are not accounted for, diagnostics are not accounted for

23 Disclaimer, Assumptions, Equipment Data

23.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

23.2 Assumptions Safety Requirements Specification

This safety requirements specification document is generated based on the selections the user made during the SIL selection and SIL verification activities in combination with specific safety requirements specification entries on both project and SIF level basis.

The cause and effect diagram that is created as part of the SIF Functional Relationship only depicts the actions to be taken for the specific SIF under consideration. If multiple SIFs initiate based on a specific sensor group and/or operate the same final element group this will not be reflected in these individual cause and effect diagrams. A complete cause and effect diagram taking into consideration all Safety Instrumented Functions will show these commonalities assuming that the user has correctly identified identical groups and has used the reuse feature in the SILver tool to identify these identical groups. A complete, i.e. project level, cause and effect diagram can be generated independent of this report.

The position of this safety requirements specification document within the overall safety lifecycle deviates from the lifecycles published in the functional safety standards. Typically the SRS phase is located between SIL selection and Conceptual Design, i.e. SIL verification. The required information in the SRS however covers information developed in the SIL selection phase as well as in the Conceptual Design / SIL verification phase. For example specific application level diagnostic requirements like external comparison of analog signals or the implementation of partial valve stroke testing are determined during the SIL verification but also need to be documented in the safety requirements specification document. Therefore the exSILentia® software allows users to create Process SRS as well as a Design SRS document to better conform to normal work practices.